SOCIOL 723: Social Statistics II

Week 10, Panel Data and Difference-in-Differences

Wenhao Jiang
Department of Sociology, Fall 2026

Duke
UNIVERSITY

Today

Panel Data and Unobserved Heterogeneity

Fixed or Random Intercepts, Without Time

Difference-in-Differences

The Trouble with Two-Way Fixed Effects

Beyond a Binary, Absorbing Treatment

Applications in Sociology

Practice

★ = essential, you must understand this ★ = for understanding, not examined

A Third Source of Variation

So far: adjust for observables (Week 6), find an instrument (Week 7), find a threshold (Week 9). Today: use variation over time within the same unit.

The core idea

Observe the same unit before and after treatment.

- What is **stable** about the unit (ability, upbringing) is the same in both periods, so the before–after change removes it, measured or not: **fixed effects**.
- What **changes for everyone** (a recession, getting older) does not cancel, but units that were not treated went through it too, so their change measures it: **difference-in-differences**.

One idea, one assumption: nothing else changes differently for the treated units.

Panel Data

The Example: Does Marriage Raise Men's Wages? ★

The question. Married men earn more than unmarried men. Is it marriage (specialization at home, employers' favour), or do men with high earning potential marry?

The data. NLSY men observed every year 1980–87: 545 men, 4,360 person-years (Vella and Verbeek 1998; `wooldridge::wagepan`). Y_{it}: log hourly wage of man i in year t; $D_{it} = 1$ if married that year.

men never married in the panel	162
men married in all eight years	73
men whose status changes	310
share married, 1980 → 1987	0.19 → 0.61
mean log wage, 1980 → 1987	1.39 → 1.87

Start simple: 1980 and 1987 only; the 528 men who married or did not change (17 who left marriage set aside). All men and years come later (figs/w10_panel_slow.R).

Pooled OLS ★

Treat the 1,056 person-years (528 men, 1980 and 1987) as if they were 1,056 different men:

$$Y_{it} = a + \beta D_{it} + u_{it}, \qquad \hat{\beta} = 0.358 \ (0.031)$$ married men earn about 43% more

$$\operatorname{plim} \hat{\beta} = \beta + \frac{\operatorname{Cov}(D_{it}, u_{it})}{\operatorname{Var}(D_{it})}$$ omitted variables (Week 2)

- u_{it} holds everything else that sets man i's wage in year t:
  - *stable* traits: ability, ambition, schooling, family background;
  - what *changes over time*: experience, seniority, prices, the economy.
- Both can move with marriage: men with high-wage traits may marry more, and men marry as they age while every man's wage grows. Either way $\operatorname{Cov}(D_{it}, u_{it}) > 0$ and 0.358 is too big.
- We cannot measure all of them. But we see each man more than once.

Standard errors clustered by man throughout (figs/w10_panel_slow.R).

Compare Each Man With Himself ★

Split the error in three: $u_{it} = \alpha_i + \lambda_t + \epsilon_{it}$.

- α_i: the combined effect on his wage of *everything stable* about man i (ability, ambition, schooling, family background), measured or not.
- λ_t: what changes for *every* man in year t (experience, prices).
- ϵ_{it}: the rest.

Take 1987 and 1980:

$$Y_{i,87} = \alpha_i + \lambda_{87} + \beta D_{i,87} + \epsilon_{i,87} \qquad \text{1987}$$

$$Y_{i,80} = \alpha_i + \lambda_{80} + \beta D_{i,80} + \epsilon_{i,80} \qquad \text{1980}$$

$$Y_{i,87} - Y_{i,80} = \underbrace{(\lambda_{87} - \lambda_{80})}_{\text{intercept}} + \beta\,(D_{i,87} - D_{i,80}) + (\epsilon_{i,87} - \epsilon_{i,80}) \qquad \text{subtract: } \alpha_i \text{ cancels}$$

First differences: regress each man's wage growth on his change in marital status, with an intercept. Idea 1: compare each man with himself. Every stable trait is gone, measured or not.

Men Who Did Not Change Are the Control Group ★

change in married, 1980 → 1987	men	mean wage growth
married ($\Delta D = 1$)	251	0.488
no change ($\Delta D = 0$)	277	0.476

OLS: $\hat{\lambda}_{87} - \hat{\lambda}_{80} = 0.476$ (0.037), $\hat{\beta} = 0.012$ (0.053).

- The intercept is the year effect, $\lambda_{87} - \lambda_{80}$, estimated from the 277 men whose status did not change. Idea 2: the year fixed effect makes them the control group.
- $\hat{\beta} = 0.488 - 0.476$ is the *extra* growth of men who married: difference-in-differences. If the timing of marriage were random, this is the comparison an experiment would make.
- Pooled OLS said 0.358. Most of it was stable traits and common growth.

All Men, All Years: The Model ★

Now all 545 men (the 17 who left marriage too) and every year, $t = 1980, \ldots, 1987$:

$$Y_{it} = \underbrace{\alpha_i}_{\text{who he is}} + \underbrace{\lambda_t}_{\text{what year it is}} + \underbrace{\beta D_{it}}_{\text{marriage}} + \underbrace{\epsilon_{it}}_{\text{everything else}} .$$

- α_i: one number per man, for everything stable about him (ability, ambition, family background), measured or not: man fixed effects.
- λ_t: one number per year, for what happens to every man that year (the economy, a year more experience): year fixed effects.
- With both sets of fixed effects in the model, people call it two-way fixed effects (TWFE).

Man Fixed Effects: Each Man His Own Line

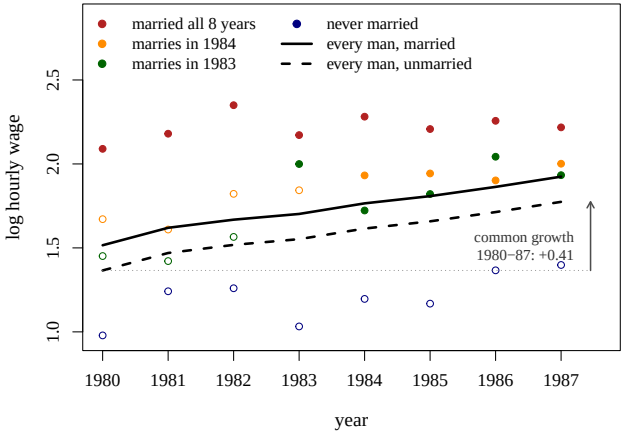
$Y_{it} = \alpha_i + \beta D_{it} + \epsilon_{it}$: one dummy per man, one common slope. Four of the men:



- The pooled line (slope 0.66 for these four) mostly compares *different* men: the high-wage men are the married ones.
- Each colored line is one man at his own height $\hat{\alpha}_i$. Its slope (0.31 for these four) compares him only with himself: married years vs. unmarried years.
- All 545 men: $\hat{\beta} = 0.243$ (0.022). But a man's married years are also his later years, when every wage is higher.

Year Fixed Effects: One Common Year Path

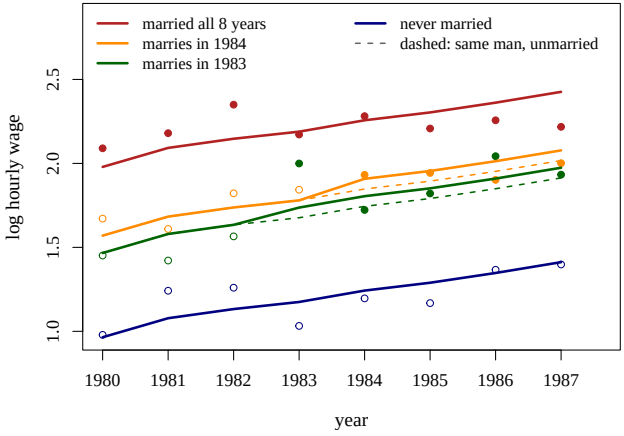
$Y_{it} = a + \lambda_t + \beta D_{it} + \epsilon_{it}$: one dummy per year, the same for every man.



- Dashed: $\hat{a} + \hat{\lambda}_t$, the wage of an unmarried man in year t. It rises 0.41 from 1980 to 1987: growth every man gets, married or not.
- Solid: the same path plus $\hat{\beta} = 0.150$ (0.028) in married years.
- But every man shares one line: the red man is high in every year, married or not, so $\hat{\beta}$ still compares different men.

Two-Way FE: Own Level, Common Year Path ★

$Y_{it} = \alpha_i + \lambda_t + \beta D_{it} + \epsilon_{it}$. Fitted on all 545 men; four of them shown:



- Every man follows the same year path, shifted to his own level.
- Marriage lifts him off his own path: the gap between solid and dashed.
- $\hat{\beta} = 0.061$ (0.022). Filled points: married years.

Two-Way FE Can Be Demeaned: the Algebra ★

Subtract the man's mean *and* the year's mean. In a balanced panel:

$$
\begin{aligned}
Y_{it} &= \alpha_i + \lambda_t + \beta D_{it} + \epsilon_{it} && \text{the model} \\
\bar{Y}_{i\cdot} &= \alpha_i + \bar{\lambda} + \beta \bar{D}_{i\cdot} + \bar{\epsilon}_{i\cdot} && \text{average over } t\text{: man } i\text{'s mean} \\
\bar{Y}_{\cdot t} &= \bar{\alpha} + \lambda_t + \beta \bar{D}_{\cdot t} + \bar{\epsilon}_{\cdot t} && \text{average over } i\text{: year } t\text{'s mean} \\
\bar{Y} &= \bar{\alpha} + \bar{\lambda} + \beta \bar{D} + \bar{\epsilon} && \text{average over both} \\
Y_{it} - \bar{Y}_{i\cdot} - \bar{Y}_{\cdot t} + \bar{Y} &= \beta(D_{it} - \bar{D}_{i\cdot} - \bar{D}_{\cdot t} + \bar{D}) + \ddot{\epsilon}_{it} && \text{line } 1 - 2 - 3 + 4\text{: } \alpha_i \text{ and } \lambda_t \text{ cancel}
\end{aligned}
$$

- **Double demeaning**: subtract the man's mean and the year's mean, add back the grand mean (subtracted twice). Same $\hat{\beta}$ as man and year dummies: 0.0606 both ways.
- With only 1980 and 1987 (528 men) it gives 0.012, the same as first differences.

How Software Does It: Dummies, or Means

Man 1 marries in 1982; man 2 never does. Omitted dummies: man 2 and 1980.

The data

man	year	Y	D	man 1	1981	1982
1	1980	1.5	0	1	0	0
1	1981	1.8	0	1	1	0
1	1982	2.4	1	1	0	1
2	1980	1.2	0	0	0	0
2	1981	1.5	0	0	1	0
2	1982	1.8	0	0	0	1

The man's mean of each column

Y	D	man 1	1981	1982
1.9	$\frac{1}{3}$	1	$\frac{1}{3}$	$\frac{1}{3}$
1.9	$\frac{1}{3}$	1	$\frac{1}{3}$	$\frac{1}{3}$
1.9	$\frac{1}{3}$	1	$\frac{1}{3}$	$\frac{1}{3}$
1.5	0	0	$\frac{1}{3}$	$\frac{1}{3}$
1.5	0	0	$\frac{1}{3}$	$\frac{1}{3}$
1.5	0	0	$\frac{1}{3}$	$\frac{1}{3}$

- Subtract right from left, row by row: the man 1 column becomes all 0, so subtracting means does the man dummies' job.
- What is left of Y is each man's distance from his own average: TWFE compares each man with himself over time.
- Man 2 gives the year path (+0.3, +0.6); man 1 grew 0.9: $\hat{\beta} = 0.3$.

What Fixed Effects Averages: One Man's Own Estimate

Man effects only, binary D. Run the within regression for *one* man:

$$b_i = \text{(his mean wage in married years)} - \text{(his mean wage in unmarried years)},$$

because a regression on one 0/1 variable with an intercept is a difference in means (Week 1). **Example** (hypothetical): married in 3 of 8 years.

	log wages	mean
5 unmarried years	1.50, 1.55, 1.60, 1.62, 1.63	1.58
3 married years	1.70, 1.75, 1.80	1.75
b_i		$1.75 - 1.58 = 0.17$

- A man who never changes status has no b_i: he adds nothing.
- Fixed effects combines the 310 men's b_i into one number. The question is how it weights them.

The Weights, Step 1: Fixed Effects Is an Average of the b_i

Tilde: minus the man's own mean. Write man i's piece of the fixed-effects formula as $N_i = \sum_t \tilde{D}_{it}\tilde{Y}_{it}$ (numerator) and $S_i = \sum_t \tilde{D}_{it}^2$ (denominator):

$$\hat{\beta}_{FE} = \frac{\sum_i \sum_t \tilde{D}_{it}\tilde{Y}_{it}}{\sum_i \sum_t \tilde{D}_{it}^2} = \frac{N_1 + N_2 + \ldots}{S_1 + S_2 + \ldots} \qquad \text{add up man by man}$$

$$b_i = N_i/S_i, \quad \text{so } N_i = S_i\, b_i \qquad \text{man } i\text{'s own regression}$$

$$\hat{\beta}_{FE} = \frac{S_1 b_1 + S_2 b_2 + \ldots}{S_1 + S_2 + \ldots} = \sum_i w_i\, b_i, \qquad w_i = \frac{S_i}{S_1 + S_2 + \ldots} \qquad \text{substitute}$$

- Fixed effects is a weighted average of the men's own estimates b_i.
- Man i's weight is his S_i, the variation in his own marriage status, as a share of everyone's.

The Weights, Step 2: What a Man's S_i Is

Man i is married in m_i of T years, so $\bar{D}_i = m_i/T = p_i$. His demeaned 0/1 values:

$$\tilde{D}_{it} = 1 - p_i \qquad \text{in his } m_i \text{ married years}$$

$$\tilde{D}_{it} = -p_i \qquad \text{in his } T - m_i \text{ unmarried years}$$

$$S_i = m_i(1-p_i)^2 + (T-m_i)\,p_i^2 \qquad \text{square and add}$$

$$= Tp_i(1-p_i)^2 + T(1-p_i)\,p_i^2 = T\,p_i(1-p_i) \qquad m_i = Tp_i\text{; factor out } Tp_i(1-p_i)$$

Check: $T = 8$, married 2 years, $p = 0.25$:
$2(0.75)^2 + 6(0.25)^2 = 1.125 + 0.375 = 1.5 = 8(0.25)(0.75)$.

years married, of 8	1	2	3	4	5	6	7
weight per man, $8p(1-p)$	0.88	1.50	1.88	2.00	1.88	1.50	0.88
share of total weight	.08	.11	.19	.15	.22	.14	.12

Largest for men married half the time; a man married 1 of 8 years counts less than half as much.

The Weights, Step 3: Two Men

	married	own estimate b_i	weight $S_i = 8p(1-p)$
man A	4 of 8 years	0.10	2.000
man B	1 of 8 years	0.40	0.875

$$\hat{\beta}_{FE} = \frac{2.000 \times 0.10 + 0.875 \times 0.40}{2.000 + 0.875} = \frac{0.55}{2.875} = 0.19$$

$$\text{plain average} = \frac{0.10 + 0.40}{2} = 0.25$$

- Fixed effects leans toward man A, whose status varies more, not toward the typical man. If effects differ across men, FE is not the average effect.

man effects only: $\hat{\beta}_{FE} = 0.243$ = weighted mean of 310 men's b_i (unweighted 0.255). Adding year effects: 0.061; wages rise with age for everyone, and marriage comes with age (figs/w10_panel.R). Toy numbers (figs/w10_toy_staggered.R).

In the Literature: Budig and England (2001)

The question. Why do mothers earn less? Does motherhood lower wages, or do women with lower earning potential have more children?

The data. NLSY79 women 1982–93: 5,287 women, 41,842 person-years.

Their model (p. 213), "effects are fixed for years and persons":

$$\ln w_{it} = \beta\, \text{children}_{it} + X'_{it}\gamma + u_i + v_t + e_{it}$$

- children_{it}: number of children (0, 1, 2, ...), not a 0/1 treatment.
- u_i: person effects (aptitude, preferences, early socialization); v_t: year effects.
- X_{it} added in blocks: marital status; human capital (education, seniority, experience, breaks in employment); job characteristics.
- No age: person and year effects already fix it (age = year − birth year).

Budig and England: Results

Each row adds a block of controls; person and year effects in every fixed-effects model.

penalty per child, their Table 2	fixed effects	pooled OLS
person and year effects only	−0.068 (0.004)	−0.081 (0.002)
+ marital status	−0.073	−0.081
+ experience, seniority, education	−0.047	−0.018
+ job characteristics	−0.037	−0.012

- About 7% per child within women, close to OLS: little selection of lower-wage women into motherhood on fixed traits.
- Experience (lost years at work) explains about a third of the FE penalty; the rest remains. Note that experience is *post*-birth, a mediator: controlling for it answers a different question.

What Two-Way FE Assumes ★

The model, with the textbook's three assumptions (Wooldridge 2010, ch. 10):

$$Y_{it} = \beta D_{it} + \underbrace{\alpha_i}_{\text{stable: the man}} + \underbrace{\lambda_t}_{\text{common: the year}} + \underbrace{\epsilon_{it}}_{\text{the rest}} .$$

	assumption	needed for	marriage data
FE.1	strict exogeneity: $E[\epsilon_{it} \mid D_{i1}, \ldots, D_{iT}, \alpha_i] = 0$, all t	unbiased, consistent	tested below
FE.2	D_{it} varies over time within some units	identification	310 of 545 men
FE.3	$\text{Var}(\epsilon_{it} \mid D_{i1}, \ldots, D_{iT}, \alpha_i) = \sigma^2$ and $\text{Cov}(\epsilon_{it}, \epsilon_{is} \mid D_{i1}, \ldots, D_{iT}, \alpha_i) = 0$, $t \neq s$	usual s.e.	false: cluster by man

- Nothing is assumed about α_i: it may correlate with D_{it} in any way.
- Random effects adds $\text{Cov}(\alpha_i, D_{it}) = 0$, all t: men who marry, early or late, have the same stable traits on average as men who do not.

Strict Exogeneity, in Words ★

Two-way FE compares each man's married years with his *own average over all years*. So FE.1 asks: is a man's marriage in any year unrelated to his wage shocks in *every* year, past, present, and future?

- **Feedback** breaks it. A raise in 1983 ($\epsilon_{i,83}$ high) makes a man more likely to marry in 1984. The raise sits in his *unmarried* years and lifts his own baseline, so FE understates the effect; if men marry after a bad year, FE overstates it.
- **Something else changing at the same time** breaks it. A man who moves for a better job and marries that year: FE credits marriage with the new job.
- What it does *not* require: anything about α_i. Stable traits may be related to marriage in any way.

Feedback leaves a trace we can test: future marriage predicting today's wage.

Strict Exogeneity, Step by Step: The Estimator's Error ★

Man effects only, one regressor, T years. Tilde: minus the man's own mean.

$$
\begin{aligned}
Y_{it} &= \alpha_i + \beta D_{it} + \epsilon_{it} && \text{the model} \\
\bar{Y}_i &= \alpha_i + \beta \bar{D}_i + \bar{\epsilon}_i && \text{average over man } i\text{'s } T \text{ years} \\
\tilde{Y}_{it} &= \beta \tilde{D}_{it} + \tilde{\epsilon}_{it} && \text{line 1} - \text{line 2: } \alpha_i \text{ cancels} \\
\hat{\beta}_{FE} &= \frac{\sum_i \sum_t \tilde{D}_{it} \tilde{Y}_{it}}{\sum_i \sum_t \tilde{D}_{it}^2} && \text{OLS on the demeaned data} \\
&= \beta + \frac{\sum_i \sum_t \tilde{D}_{it} \tilde{\epsilon}_{it}}{\sum_i \sum_t \tilde{D}_{it}^2} && \text{substitute line 3 for } \tilde{Y}_{it}
\end{aligned}
$$

- Unbiased if the last fraction averages to zero: $\tilde{D}_{it}$ unrelated to $\tilde{\epsilon}_{it}$.
- But $\tilde{\epsilon}_{it} = \epsilon_{it} - \bar{\epsilon}_i$, and $\bar{\epsilon}_i = (\epsilon_{i1} + \cdots + \epsilon_{iT})/T$ holds *every* year's shock. So marriage in one year meets shocks from all years.

Testing Strict Exogeneity: Add the Lead

Under FE.1, next year's treatment cannot predict this year's outcome once α_i and this year's treatment are held fixed:

$$Y_{it} = \beta D_{it} + \delta D_{i,t+1} + X_{it}'\gamma + \alpha_i + \lambda_t + \epsilon_{it}, \qquad H_0 : \delta = 0.$$

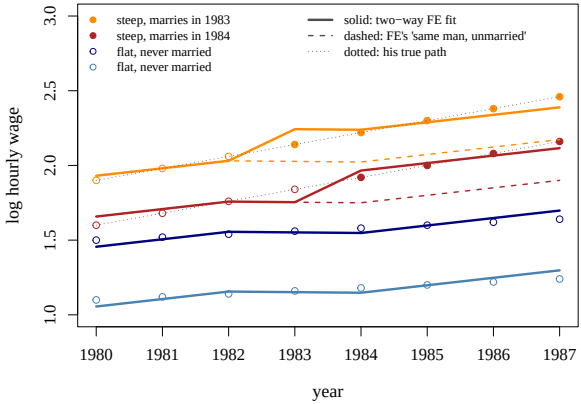
marriage data, 1980–86 (lead needs 1987)	estimate	(se)
married this year	0.048	(0.025)
married next year	−0.0005	(0.022)

- No sign that men marry after a wage jump: $p = 0.98$.
- The s.e. (0.022) is as large as the effect: a small feedback would go unseen.
- This is the first pre-period coefficient of an event study (DiD section).

Man and year effects, union, experience^2; clustered by man (figs/w10_panel_std.R).

When Men on Steeper Paths Marry

Hypothetical, no noise, and marriage has no effect at all. Fitted by two-way FE:



- Steep wage paths (0.08 a year) marry; flat paths (0.02) never do.
- α_i: each man his own *level*. λ_t: one *common* year path. No man gets his own growth.
- The common path is too flat for the steep men: after marriage they run above it, and FE calls the gap marriage, $\hat{\beta} = 0.22$.
- Parallel trends fails. Own slopes: $\hat{\beta} = 0$ (figs/w10_slopes.R).

In the Literature: Ludwig and Brüderl (2018)

The question. Is the male marriage wage premium causal, or do men on steeper careers marry? A test of specialization theory.

The data. NLSY79, 1979–2012: 4,287 employed men, 49,801 person-years; men observed never married at least once.

The design. Four estimators of one coefficient:

model	allows men to differ in	marriage premium
pooled OLS	nothing	17.7%
FE	wage level	8.3%
FE, group slopes	level; growth by future marital status	4.3%
FEIS	level and own wage growth	0.5% (n.s.)

Their Figure 3. Before marrying, future husbands' wages grew 5% per year of experience; never-married men's, 3%.

Random Trends: Each Man His Own Slope

FE allows men to differ in wage *levels*. If men on steeper wage paths marry, FE still mistakes their growth for a marriage effect. Give each man his own line:

$$Y_{it} = \alpha_i + \gamma_i\, x_{it} + \lambda_t + \beta D_{it} + \epsilon_{it} \qquad \text{own intercept } \alpha_i\text{, own slope } \gamma_i \text{ in experience } x$$

Year effects stay: own lines absorb differences in *growth* between men; λ_t absorbs shocks common to a year (a recession), which lie on no one's line.

Estimate β in three steps:

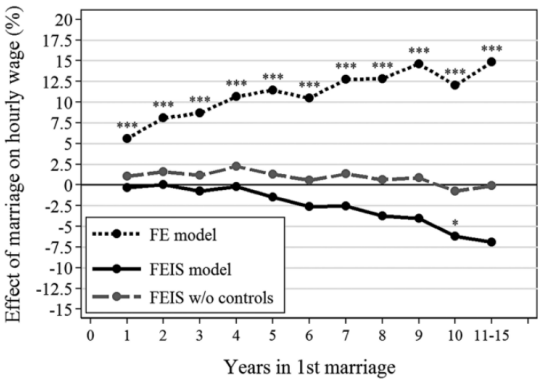
1. For each man, regress his wages on $(1, x_{it})$ using only his own years. Keep the residuals $\ddot{Y}_{it}$: his departures from his own line.
2. Do the same for marriage, giving $\ddot{D}_{it}$, and for each year dummy.
3. Pool all men and regress $\ddot{Y}_{it}$ on $\ddot{D}_{it}$ and the detrended year dummies.

Fixed effects is the same recipe with a line of slope zero: step 1 subtracts only his mean.

Random Trends: What Changes

- The **random trend model** (Wooldridge 2010, ch. 11); sociology calls it **FEIS**, fixed effects with individual slopes: detrend each unit separately, then pool (Brüderl and Ludwig 2015; Rüttenauer and Ludwig 2023). Needs at least three observations per unit.
- Identifies β from *departures* from each man's own trajectory at marriage: a jump, not a steeper path.
- Our 8-year panel: FE 0.047 (0.021), FEIS 0.055 (0.024). Eight years per man is short for estimating a slope; the NLSY79 over 34 years (next frames) tells a different story.

Ludwig and Brüderl: the Premium Over Time



Their Figure 4: effect by years since first marriage.

- FE: the premium grows to 11.4% after 5 years and 14.8% after 11–15, the pattern read as evidence of specialization.
- FEIS: about zero throughout.
- The growing "premium" is the steeper wage path the men were already on: selection on growth, not an effect of marriage.
- Generalizes: an FE impact function that keeps rising after the event is a warning sign of selection on slopes.

Two-Way Fixed Effects: Strengths and Weaknesses ★

Strengths

- Removes every stable trait of the unit, *measured or not*.
- Removes everything common to a year.
- Simple and transparent: a difference-in-differences with many units and periods.
- Testable in part: leads (no feedback), event studies.

Weaknesses

- Needs strict exogeneity: no feedback, nothing else changing at the same time.
- Misses differences in *growth*: men on steeper paths (FEIS).
- Only units whose treatment changes identify β, weighted by how much it varies.
- Less precise; measurement error grows. Cluster by unit.
- With a lagged outcome: Nickell bias (below).

Why Put Last Year's Outcome in a Fixed-Effects Model?

Marriage can follow wages, not only the other way round: a good year makes a man more likely to marry next year, and wages persist. So control for last year's wage:

$$Y_{it} = \rho\, Y_{i,t-1} + \beta D_{it} + \alpha_i + \epsilon_{it}$$

This compares men who start the year at the same wage, one who marries and one who does not; α_i still removes everything stable about the man.

Other questions where the outcome feeds back into the treatment:

- **Health and employment**: illness leads to job loss; job loss worsens health.
- **Parental help and test scores**: parents step in after a bad report card.
- **Policing and crime**: cities hire police after a crime spike; crime persists.

The catch: with α_i in the model, a lagged outcome breaks strict exogeneity.

Fixed Effects or a Lagged Outcome?

Sociologists often control for last year's outcome, to absorb differences between units that are *changing* rather than fixed. Combining it with unit fixed effects creates a new bias (Nickell, next frame).

- **FE and LDV embody different selection stories**: FE assumes selection on time-invariant traits; LDV assumes selection on prior outcomes. They can both be valid, but they are different assumptions and will generally give different answers.
- **Bracketing** (Angrist and Pischke 2009, sec. 5.4): for a positive effect, if LDV is the right model, FE overstates it; if FE is the right model, LDV understates it. Run both: the truth lies between.

Dynamic Panels: Nickell Bias

The outcome depends on its own past:

$$Y_{it} = \rho Y_{i,t-1} + X_{it}'\beta + \alpha_i + \epsilon_{it}.$$

FE.1 fails by construction: $Y_{i,t-1}$ contains $\epsilon_{i,t-1}$, which is in $\bar{\epsilon}_{i\cdot}$. Demeaned lag and demeaned error are correlated (Nickell 1981):

$$\text{plim}_{N\to\infty}\ \hat{\rho}_{FE} - \rho \ \approx\ -\frac{1+\rho}{T-1}$$

large-T approximation; exact formula in the script

simulation, $\rho = 0.5$, $N = 500$, 200 draws	$T = 4$	$T = 8$	$T = 16$
mean of $\hat{\rho}_{FE}$	-0.04	0.27	0.40
Nickell's exact fixed-T value	-0.04	0.27	0.39

Large N does not help; only large T does (figs/w10_panel_std.R).

Anderson–Hsiao: Difference, Then Instrument

First-difference to remove α_i:

$$\Delta Y_{it} = \rho\,\Delta Y_{i,t-1} + \Delta X_{it}'\beta + \Delta\epsilon_{it} \qquad \alpha_i \text{ gone}$$

$$\Delta Y_{i,t-1} = Y_{i,t-1} - Y_{i,t-2}, \quad \Delta\epsilon_{it} = \epsilon_{it} - \epsilon_{i,t-1} \qquad \text{both contain } \epsilon_{i,t-1}\text{: endogenous}$$

Instrument $\Delta Y_{i,t-1}$ with $Y_{i,t-2}$ (Anderson and Hsiao 1981), the Week 7 conditions:

- **Relevance**: $Y_{i,t-2}$ is part of $\Delta Y_{i,t-1}$.
- **Exogeneity**: $Y_{i,t-2}$ depends on errors up to $t-2$; $\Delta\epsilon_{it}$ on t and $t-1$. Holds only if ϵ_{it} is serially uncorrelated.

same simulation, $\rho = 0.5$	$T = 4$	$T = 8$	$T = 16$
mean of $\hat{\rho}_{AH}$ (sd)	0.51 (0.13)	0.51 (0.05)	0.50 (0.03)

Arellano–Bond: Use Every Valid Lag

If ϵ_{it} is serially uncorrelated, *every* lag from $t-2$ back is a valid instrument for period t's differenced equation:

$$E\big[Y_{i,t-s}\,\Delta\epsilon_{it}\big] = 0, \qquad s \geq 2,\ t = 3, \ldots, T.$$

GMM combines all these moment conditions (Arellano and Bond 1991).

- More instruments, more precision: period t has $t-2$ of them. With $T = 8$ and year effects: 29 instruments.
- **The key check**: the differenced residuals $\Delta\hat{\epsilon}_{it}$. First-order correlation is expected: $\Delta\epsilon_{it}$ and $\Delta\epsilon_{i,t-1}$ share $\epsilon_{i,t-1}$. Second-order correlation must be zero; if it is not, ϵ_{it} is serially correlated and $Y_{i,t-2}$ is invalid. Start at $Y_{i,t-3}$ and check *third*-order correlation instead.
- Too many instruments overfit the first stage and pull GMM toward FE-style bias; report the count (Roodman 2009).

A Dynamic Wage Equation on the Marriage Data

log wage on its lag, married, union (both treated as strictly exogenous); year effects; robust s.e.; `fixest` (AH), `plm::pgmm` one-step (AB) (figs/w10_panel_std.R):

estimator	lagged wage $\hat{\rho}$	married	note
pooled OLS	0.60 (0.03)	0.056 (0.015)	α_i in the error: $\hat{\rho}$ too high
FE	0.06 (0.02)	0.062 (0.021)	Nickell: $\hat{\rho}$ too low
Anderson–Hsiao, Y_{t-2}	0.12 (0.03)	0.046 (0.022)	first-stage $F = 830$
Arellano–Bond, Y_{t-2} on	0.16 (0.04)	0.044 (0.023)	2nd-order test $z = 2.18$, $p = 0.03$
Arellano–Bond, Y_{t-3} on	0.68 (0.14)	0.031 (0.031)	3rd-order test $p = 0.97$
Anderson–Hsiao, Y_{t-3}	0.81 (0.27)	0.027 (0.034)	first-stage $F = 12$

- The second-order test rejects: wage shocks persist, so Y_{t-2} is not a valid instrument, and the rows using it are biased.
- Lag 3 passes the third-order test but is weak ($F = 12$): $\hat{\rho}$ of 0.7–0.8, above pooled OLS (an upper bound). Eight years do not pin down persistence.
- The marriage coefficient, 0.03–0.06, is far less sensitive than $\hat{\rho}$.

FE vs RE

Fixed or Random Intercepts: the Question

Goal: the same model whenever members share a group (students in a school, siblings in a family, workers in a firm, the years of one life):

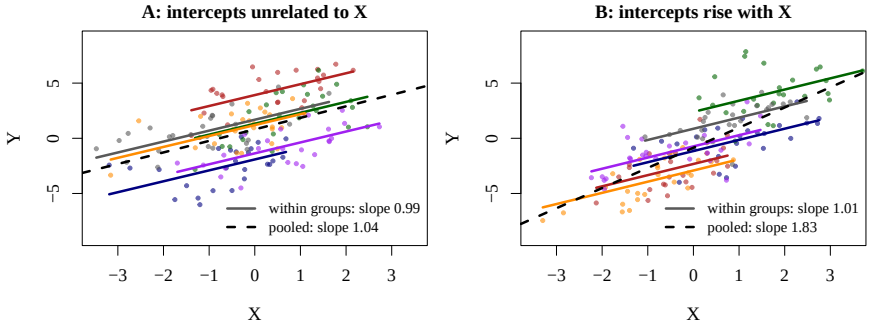
$$Y_{ij} = \alpha_j + \beta X_{ij} + \epsilon_{ij}.$$

	assumption on α_j	uses
fixed effects	none: α_j may covary with X	within groups only
random effects	$\text{Cov}(\alpha_j, X_{ij}) = 0$, all i: unrelated to X	within and between

- Both let α_j vary across groups. The names mislead: the key difference is whether α_j may covary with X (Wooldridge 2010, ch. 10).
- If it may not, random effects is more precise and estimates group-level variables.

Two Worlds in a Picture

Six groups, $\beta = 1$; same $\mathrm{Var}(\alpha_j)$ in both worlds, only its covariance with X differs (figs/w10_fe_sim.R):



- A: high-X groups are no higher; pooled and within slopes agree (1.04 vs 0.99).
- B: high-X groups have high intercepts; the pooled slope adds that gap (1.83 vs 1.01).

Three Estimators, One Formula

Start from the model, average it over the group's members, and subtract a share θ of that average:

$$Y_{ij} = \beta X_{ij} + \alpha_j + \epsilon_{ij} \qquad \text{the model}$$

$$\bar{Y}_j = \beta \bar{X}_j + \alpha_j + \bar{\epsilon}_j \qquad \text{average over group } j\text{'s members}$$

$$Y_{ij} - \theta \bar{Y}_j = \beta(X_{ij} - \theta \bar{X}_j) + (1-\theta)\,\alpha_j + (\epsilon_{ij} - \theta \bar{\epsilon}_j) \qquad \text{line 1 minus } \theta\times \text{ line 2}$$

θ	estimator	share of α_j left in the error
0	pooled OLS	all
between 0 and 1	random effects	$1-\theta$
1	fixed effects	none

- For $\theta < 1$ a share $(1-\theta)$ of α_j stays in the error. That is harmless only if α_j is unrelated to X: the random-effects assumption.

What θ Is: How Much to Trust the Group Mean

Random effects picks θ from two variances, $\tau^2 = \text{Var}(\alpha_j)$ and $\sigma^2 = \text{Var}(\epsilon_{ij})$, with n members per group:

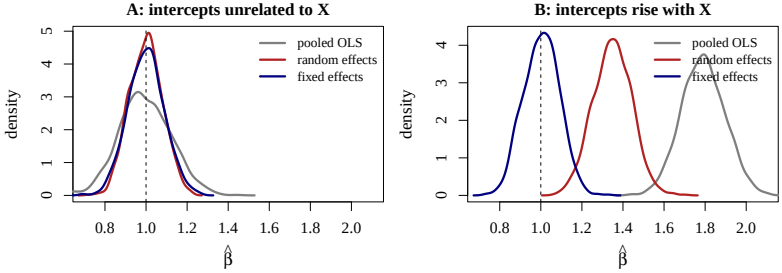
$$\theta = 1 - \sqrt{\frac{\sigma^2}{\sigma^2 + n\tau^2}} \qquad \text{(GLS; Wooldridge 2010, ch. 10).}$$

- The group mean $\bar{Y}_j$ carries α_j plus noise of variance σ^2/n.
- Many members or large group differences ($n\tau^2 \gg \sigma^2$): the group mean is mostly α_j, so remove nearly all of it, $\theta \to 1$, like fixed effects.
- Few members or small group differences: the group mean is mostly noise, so remove little of it, $\theta \to 0$, like pooled OLS.

$\sigma = 1.5$, $\tau = 2$	$n = 1$	$n = 4$ (simulation)	$n = 20$
θ	0.40	$1 - \sqrt{2.25/18.25} = 0.65$	0.84

1,000 Samples in Each World

100 groups of 4, $\beta = 1$:



mean (sd) of $\hat{\beta}$	pooled OLS	random effects	fixed effects
A: unrelated	0.995 (0.126)	0.999 (0.080)	1.000 (0.088)
B: rise with X	1.790 (0.105)	1.348 (0.094)	1.003 (0.087)

A: all unbiased, random effects most precise. B: random effects keeps $1 - \theta = 35\%$ of α_j and is biased; fixed effects is not.

Testing Fixed Against Random: Hausman

H_0: α_j is unrelated to X (Hausman 1978). Fixed effects is consistent either way; random effects is consistent and more precise under H_0, inconsistent otherwise.

$$H = \frac{(\hat{\beta}_{FE} - \hat{\beta}_{RE})^2}{\text{Var}(\hat{\beta}_{FE}) - \text{Var}(\hat{\beta}_{RE})} \xrightarrow{d} \chi^2_1$$

- **Numerator**: how far apart the two estimates are. Under H_0 both are right, so they should differ only by noise.
- **Denominator**: how much noise to expect. Under H_0 random effects is the most precise, so $\text{Var}(\hat{\beta}_{FE} - \hat{\beta}_{RE}) = \text{Var}(\hat{\beta}_{FE}) - \text{Var}(\hat{\beta}_{RE})$.
- Large H: the gap is more than noise, so reject random effects. In the 1,000 simulated samples it rejects 5.6% of the time in world A, 100% in world B.

Marriage data: $\chi^2_{10} = 37.0$, $p < 0.001$. Caveat: the denominator assumes random effects is fully efficient, which fails when errors are serially correlated.

On the Marriage Data: Random Effects

RE is OLS after subtracting a share θ of each man's mean. In a balanced panel, one θ for everyone:

$$Y_{it} - \theta \bar{Y}_{i\cdot} = (X_{it} - \theta \bar{X}_{i\cdot})'\beta + \ldots, \qquad \theta = 1 - \sqrt{\frac{\sigma_\epsilon^2}{\sigma_\epsilon^2 + T\sigma_\alpha^2}}$$

$$\hat{\theta} = 1 - \sqrt{\frac{0.123}{0.123 + 8(0.105)}} = 0.64 \qquad \text{marriage data}$$

- The man effect is 46% of the error variance; RE removes 64% of each man's mean.
- RE needs FE.1 *and* $\mathrm{Cov}(\alpha_i, X_{it}) = 0$, all t: the man effect unrelated to his whole marriage history.
- RE's 0.064 lies between pooled OLS with controls (0.108) and FE (0.047): it partly uses the between-man comparison, and so can estimate education and race.

Swamy–Arora variance components, `plm` (figs/w10_panel_std.R).

Fixed or Random: How to Choose

if the question is	use
the effect of a member-level variable, and the group effect may be confounded	fixed effects
the effect of a group-level variable (school sector)	random effects (fixed effects cannot estimate it), which needs α_j unrelated to every regressor
how much variation lies between groups (Week 4's ICC)	random effects

- Not rejecting does not make fixed effects wrong: it stays consistent either way.
- Fixed effects is not automatically causal: it removes only what group members share.

DID

The Canonical 2×2 ★

Two groups, two periods. One group is treated in period 2.

	Before	After	Difference
Treated	$E[Y \mid g{=}1, t{=}1]$	$E[Y \mid g{=}1, t{=}2]$	Δ_1
Control	$E[Y \mid g{=}0, t{=}1]$	$E[Y \mid g{=}0, t{=}2]$	Δ_0
			$\tau = \Delta_1 - \Delta_0$

- The first difference removes anything fixed about each group.
- The second difference removes anything common to both groups over time.
- Equivalent regression: $Y_{it} = \alpha + \gamma\,\text{treated}_i + \delta\,\text{post}_t + \tau\,(\text{treated}_i \times \text{post}_t) + \epsilon_{it}$.
- Example from the panel section: wage growth 1980–87 was 0.488 for men who married and 0.476 for men who did not change, so $\tau = 0.012$.

The 2×2 in the Literature: Card and Krueger (1994)

New Jersey raised its minimum wage from \$4.25 to \$5.05 in April 1992; Pennsylvania did not. 410 fast-food restaurants surveyed in February and November 1992.

FTE employment per store (se), their Table 3	PA (79)	NJ (331)	NJ − PA
February (before)	23.33 (1.35)	20.44 (0.51)	−2.89 (1.44)
November (after)	21.17 (0.94)	21.03 (0.52)	−0.14 (1.07)
change	−2.16 (1.25)	0.59 (0.54)	2.76 (1.36)

- $2.76 = 0.59 - (-2.16)$: Pennsylvania's decline stands in for what New Jersey would have done without the law.
- Parallel trends here is one claim about one region in one season. Their within-NJ check: high-wage NJ stores, not bound by the law, changed like PA (−2.16 vs. −2.28, balanced sample).

The 2×2 in Regression Form

Group $g_i \in \{0, 1\}$, period $t \in \{1, 2\}$; the four cell means follow:

$$Y_{it} = \alpha + \gamma\, g_i + \delta\, \text{post}_t + \tau\, (g_i \times \text{post}_t) + \epsilon_{it}.$$

	Before	After	Difference
Treated ($g = 1$)	$\alpha + \gamma$	$\alpha + \gamma + \delta + \tau$	$\delta + \tau$
Control ($g = 0$)	α	$\alpha + \delta$	δ
			τ

- γ absorbs fixed differences in level; δ absorbs the common time trend; τ is what is left.
- Card and Krueger: $\alpha = 23.33$ (PA, February), $\gamma = -2.89$ (NJ minus PA in February), $\delta = -2.16$ (PA's change), $\tau = 2.76$.
- Unit and period fixed effects in place of g_i and post_t: two-way FE, identical with two groups and two periods, not otherwise.

Parallel Trends ★

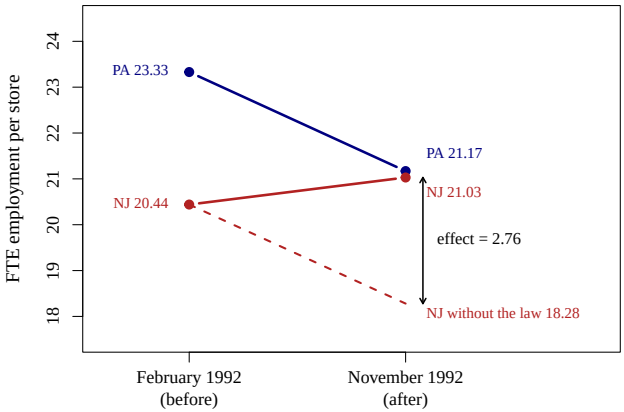
The identifying assumption

$$E[Y_{i2}(0) - Y_{i1}(0) \mid g_i = 1] = E[Y_{i2}(0) - Y_{i1}(0) \mid g_i = 0].$$

Absent treatment, the two groups' average outcomes would have moved in parallel.

- Levels may differ arbitrarily. Only the *trends* in the untreated potential outcome must agree.
- Untestable: it concerns a counterfactual trend. Pre-treatment parallelism is suggestive evidence, not proof.
- **Not scale-free**: parallel trends in levels and parallel trends in logs are different assumptions, and both cannot generally hold.
- Needs **no anticipation**: units must not change behaviour before treatment in expectation of it.

Parallel Trends in a Picture: Card and Krueger



- The assumption: without the law, NJ stores would have lost 2.16 workers, as PA stores did.
- Dashed: $20.44 - 2.16 = 18.28$, NJ without the law.
- Effect: $21.03 - 18.28 \approx 2.76$.
- Levels may differ (NJ started 2.89 lower); only the change must match.

Their Table 3; 2.75 from the rounded means (figs/w10_ck_parallel.R).

What the Double Difference Identifies, Exactly ★

Decompose the treated group's observed change. Its period-2 outcome is $Y_{i2}(1)$, its period-1 outcome is $Y_{i1}(0)$ (no anticipation); add and subtract the counterfactual $Y_{i2}(0)$:

$$\Delta_1 = E[Y_{i2}(1) - Y_{i1}(0) \mid g_i{=}1] = \underbrace{E[Y_{i2}(1) - Y_{i2}(0) \mid g_i{=}1]}_{\text{ATT}} + \underbrace{E[Y_{i2}(0) - Y_{i1}(0) \mid g_i{=}1]}_{\text{treated group's } \textit{untreated} \text{ trend}}.$$

The control group supplies an estimate of that second, unobservable term: $\Delta_0 = E[Y_{i2}(0) - Y_{i1}(0) \mid g_i{=}0]$. Subtracting,

$$\boxed{\Delta_1 - \Delta_0 \;=\; \text{ATT} \;+\; \underbrace{E[\Delta Y(0) \mid g{=}1] - E[\Delta Y(0) \mid g{=}0]}_{\text{differential trend; } = 0 \text{ under parallel trends}}}$$

Reading the Double Difference ★

$\Delta_1 - \Delta_0 = \text{ATT} + \text{differential trend}$, so:

- The estimand is the ATT, not the ATE: we borrowed a counterfactual trend *for the treated group*. Nothing in the design identifies what treatment would have done to the controls.
- The bias term *is* the parallel-trends violation, exactly. Every diagnostic in this lecture (pre-trends, placebos, Rambachan–Roth bounds) is aimed at that one object.
- Card and Krueger: NJ's change $0.59 = \text{ATT} +$ NJ's untreated trend. PA says that trend is -2.16, so ATT $= 2.76$. Had NJ's untreated trend really been -1.00, the DiD would still say 2.76 but the ATT would be $0.59 + 1.00 = 1.59$: the bias term is $-1.00 - (-2.16) = 1.16$.

Parallel Trends Is Not Scale-Free ★

Suppose untreated outcomes would have grown by 10% in both groups, from a base of 20 in the treated group and 40 in the control group.

	Before	After	Change
Treated	20	22	+2
Control	40	44	+4

- Parallel trends holds in logs (both +10%) but fails in levels (+2 versus +4). A DID in levels would report a spurious −2.
- Here both hold only if base levels are equal or nothing grows. There is no neutral choice: the transformation is part of the identifying assumption.
- Choose before looking at results: would untreated outcomes grow by the same amount, or the same percentage? It matters most when levels are far apart.

When Does the Scale Not Matter? (Roth and Sant'Anna 2023a)

Parallel trends in levels, in logs, in ranks: when can all hold at once?

Result

Parallel trends holds for *every* monotone transformation of Y if and only if the whole *distribution* of $Y(0)$ satisfies a parallel-trends condition. In practice: the population splits into a part where treatment is as good as random and a part whose $Y(0)$ distribution does not change over time.

- In any real DiD the scale is therefore part of the assumption. Choose it before seeing results, on substantive grounds (previous frame).
- The condition can be falsified in the pre-period: compare the groups' outcome distributions, not only means.
- For effects on whole distributions under weaker assumptions: Callaway and Li (2019) (later frame).

Why Control for Covariates in a DiD?

Fixed effects already remove every *level* difference between units. A covariate X matters only if it predicts the *trend*:

- treated and control units differ in X, and
- units with different X would have changed differently even without treatment.

Then parallel trends fails for the groups as a whole but can hold *within* levels of X:

$$E[\Delta Y(0) \mid X = x,\, D = 1] = E[\Delta Y(0) \mid X = x,\, D = 0] \qquad \text{for each } x.$$

Examples

- **Minimum wage**: NJ and PA have different mixes of store types, and the types were shrinking at different rates.
- **Job training**: trainees are younger, and younger workers' wages grow faster anyway.
- **Marriage**: college men marry later and have steeper wage growth.

Why Condition on X: A Worked Example

Hypothetical: two kinds of stores. Without the law, type A loses 3 workers and type B loses 1. The law adds 2 to every NJ store. NJ is mostly type B, PA mostly type A.

	share in NJ	share in PA	NJ change	PA change	DiD
type A	0.2	0.8	$-3+2=-1$	-3	2
type B	0.8	0.2	$-1+2=\ \ 1$	-1	2
all stores			0.6	-2.6	3.2

- NJ all: $0.2(-1)+0.8(1)=0.6$. PA all: $0.8(-3)+0.2(-1)=-2.6$.
- The plain DiD says 3.2, not 2: PA declines faster because it has more type A stores. Parallel trends fails for all stores but holds within each type.
- Compare within type, then average with NJ's shares: $0.2(2)+0.8(2)=2$. That is the regression estimator; weighting and doubly robust do the same.

Estimator 1, Regression: Predict NJ's Untreated Change

Same hypothetical stores. Learn the untreated change by type from PA, then predict it for NJ:

1. From PA stores: untreated change $m(\text{A}) = -3$, $m(\text{B}) = -1$.
2. Predict each NJ store's untreated change from its type, and average over NJ: $0.2(-3) + 0.8(-1) = -1.4$.
3. ATT = NJ's actual change − predicted untreated change $= 0.6 - (-1.4) = 2.0$.

This is g-computation (Week 6) on the change ΔY; with discrete X and one cell per value, it is cell-by-cell matching. In symbols, with $m_{0,\Delta}(x) = E[\Delta Y \mid X = x, D = 0]$:

$$\text{ATT} = E[\Delta Y_i \mid D_i = 1] - E[m_{0,\Delta}(X_i) \mid D_i = 1].$$

Estimator 2, Weighting: Make PA Look Like NJ

Say 100 stores in each state: type A is 20 NJ and 80 PA stores, type B is 80 NJ and 20 PA.

1. Propensity score: share in NJ within each type, $e(\text{A}) = 20/100 = 0.2$, $e(\text{B}) = 0.8$.
2. Weight each PA store by $e/(1-e)$: type A $0.2/0.8 = 0.25$, type B $0.8/0.2 = 4$.
3. Weighted PA now has $80 \times 0.25 = 20$ type A and $20 \times 4 = 80$ type B stores: NJ's mix.
4. Weighted PA change $= (20 \cdot (-3) + 80 \cdot (-1))/100 = -1.4$, so ATT $= 0.6 - (-1.4) = 2.0$.

Doubly robust combines the two: right if either the predictions m or the weights e are right.

Parallel Trends Given Covariates: Three Estimators

Assume parallel trends only within levels of pre-treatment X; two periods, $\Delta Y_i = Y_{i2} - Y_{i1}$, $e(x) = \Pr(D_i = 1 \mid X_i = x)$, $m_{0,\Delta}(x) = E[\Delta Y_i \mid X_i = x, D_i = 0]$:

$$\text{regression:} \quad \text{ATT} = E[\Delta Y_i \mid D_i = 1] - E[m_{0,\Delta}(X_i) \mid D_i = 1]$$

$$\text{weighting:} \quad \text{ATT} = E\Big[\frac{D_i - e(X_i)}{\Pr(D=1)\,(1 - e(X_i))}\,\Delta Y_i\Big]$$

$$\text{doubly robust:} \quad \text{ATT} = E\big[(w_1(D_i) - w_0(D_i, X_i))\,(\Delta Y_i - m_{0,\Delta}(X_i))\big]$$

with $w_1 = D/E[D]$ and $w_0 \propto e(X)(1-D)/(1-e(X))$, normalized to mean one (Heckman, Ichimura, and Todd 1997; Abadie 2005; Sant'Anna and Zhao 2020).

- This is Week 6 with ΔY as the outcome: compare treated and control units with the same pre-treatment X. Needs overlap, $e(x) < 1$.
- The DR version is consistent if either e or $m_{0,\Delta}$ is right (`DRDID`); it is what `did` uses inside each $ATT(g,t)$.

Covariates in TWFE: What Works

Same hypothetical stores, 100 per state, two periods (figs/w10_did_cov.R):

	effect +2 for both types	effect A +4, B +1
true ATT	2.00	1.60
TWFE	3.20	2.80
TWFE + type dummy	3.20	2.80
TWFE + type × period effects	2.00	2.50
regression (g-computation)	2.00	1.60

- A type dummy does nothing: type never changes, so the store effects absorb it.
- Type × period effects give each type its own year path: the TWFE way to compare within type. Right when the effect is the same for every type.
- When it differs, TWFE weights each type by its treatment variance (here equal, so $(4+1)/2 = 2.5$); the ATT weights by NJ's mix: $0.2 \times 4 + 0.8 \times 1 = 1.6$.

Why Not Just Add Covariates to TWFE?

$Y_{it} = \alpha_i + \lambda_t + \beta D_{it} + X'_{it}\delta + \epsilon_{it}$ is the default. Even with two periods it fails when (Caetano and Callaway 2024):

1. X_{it} is affected by treatment (a "bad control", Week 5). Example: hours per worker after a minimum-wage rise;
2. trends depend on the *level* of X, not only its change: the within transformation keeps only changes. Example: large stores shrink faster;
3. trends depend on time-invariant X (region, education), which the unit effects absorb, unless interacted with period (previous table);
4. effects vary with X, so the regression weights the subgroups in its own way (Week 6's regression weights again);
5. the linear form is wrong.

Fix: use pre-treatment X (or X at a baseline period) in one of the three estimators on the previous frame; never post-treatment X.

Event Studies ★

Many periods: one effect for each period relative to treatment:

$$Y_{it} = \alpha_i + \lambda_t + \sum_{k \neq -1} \tau_k \cdot \mathbb{1}\{t - G_i = k\} + \epsilon_{it},$$

where G_i is unit i's treatment date (for example, the year a man marries) and $k = -1$ is the omitted reference.

- **Pre-period coefficients** ($k < 0$) are the parallel-trends check. Flat and near zero is what you want to see.
- **Post-period coefficients** trace out the dynamics: does the effect grow, fade, or appear only after a lag?
- Caution: a flat pre-trend is weak evidence. The test often has low power, and conditioning your analysis on passing it distorts inference (Roth 2022).
- With staggered G_i and effects that differ by cohort, these TWFE leads are contaminated by post-period effects (Staggered section).

Reading the Event-Study Equation

One man married in 1984 ($G_i = 1984$), observed 1980–87. $k = t - G_i$ is years since marriage:

year t	1980	1981	1982	1983	1984	1985	1986	1987
$k = t - 1984$	-4	-3	-2	-1	0	1	2	3
dummy that is 1	τ_{-4}	τ_{-3}	τ_{-2}	none	τ_0	τ_1	τ_2	τ_3

- Each year switches on exactly one dummy, except $k = -1$, the reference: every τ_k compares year k with the year before marriage.
- A man who never marries has every dummy 0: he only helps estimate α_i and λ_t.
- $\tau_{-4}, \tau_{-3}, \tau_{-2}$: marriage cannot have caused anything yet, so they should be near 0.

Event Studies: Specification Details That Matter

Three choices inside the specification change the picture; report each.

- **Normalization**: one lead must be omitted (here $k = -1$). Every τ_k is relative to it, so the picture shifts if you change the reference period. Say which you used.
- **Binning the endpoints**: with unbalanced exposure, far leads and lags are estimated off few units. Bin them ($k \leq -5$, $k \geq 5$) or restrict to a balanced window, and say which.
- **Never-treated units** anchor the time effects λ_t. If there are none, the fully dynamic TWFE specification is not identified (a linear trend in k is indistinguishable from $\alpha_i + \lambda_t$): drop a second lead or bin (Borusyak, Jaravel, and Spiess 2024).

Pre-Trend Tests Have Low Power (Roth 2022)

The practice: estimate the leads, see that they are insignificant, proceed as if parallel trends holds.

The audit: 12 event-study papers in leading economics journals (2014–18) with replication data. Roth asks what happens under a linear violation of parallel trends that the usual pre-test would catch only half the time.

- The bias such a violation produces is often as large as the estimated effect, sometimes larger.
- In the worst case a nominal 95% interval covers the truth 24% of the time.
- **Conditioning on passing makes it worse**: among draws that pass the pre-test, the bias is usually larger, sometimes twice as large, because passing selects draws whose noise happened to offset the trend.

Practical: report the power of the pre-test against violations you care about (`pretrends`), and replace "test and proceed" with sensitivity analysis (Rambachan and Roth 2023, below).

Base Periods: Why Event-Study Plots Disagree (Roth 2026)

No effect at all; untreated outcomes of the treated group drift by γ per period relative to controls; treatment starts at $r = 0$.

$$\text{universal base } (r = -1): \quad \hat{\beta}_r = \gamma\,(r+1) \quad \text{for every } r \qquad \text{as TWFE: one straight line}$$

$$\text{varying base, } r < 0: \quad \hat{\beta}_r = \gamma \qquad \text{compares } r \text{ with } r-1$$

$$\text{varying base, } r \geq 0: \quad \hat{\beta}_r = \gamma\,(r+1) \qquad \text{compares } r \text{ with } -1$$

$\gamma = 0.1$, $r =$	-4	-3	-2	-1	0	1	2
universal base	-0.3	-0.2	-0.1	0	0.1	0.2	0.3
varying base	0.1	0.1	0.1	0.1	0.1	0.2	0.3

Base Periods: Reading the Two Plots

Same setup: no effect, the treated group drifts by $\gamma = 0.1$ per period:

- Callaway–Sant'Anna's default ("varying") draws a flat pre-period and a rising post-period: a kink at zero that looks like an effect. The imputation estimator's default draws a jump.
- The pre-period points answer different questions under the two bases. Do not read a varying-base plot with TWFE eyes, and do not feed it to `HonestDiD`.
- For plots: `base_period = "universal"` in `did`. All event studies in this lecture use it.

Drawing an Event Study (Freyaldenhoven, Hansen, Pérez Pérez, and Shapiro 2025)

- **Normalize** $\hat{\beta}_{-1} = 0$ and label that point with the outcome's mean, so readers can judge magnitudes.
- **Two kinds of bands**: pointwise 95% intervals, and a sup-t band that covers the whole path at once. A path shape ("no pre-trend at all leads") is a joint claim; only the uniform band tests it.
- **Report two p-values** in the legend: no pre-trends (leads jointly zero) and effects levelling off (last lags equal).
- **Show the least wiggly confound** consistent with the leads: if a smooth trend through the pre-period would explain the post-period, the plot should show it.
- **Bin or trim the end points** and say which.

Freyaldenhoven, Hansen, and Shapiro (2019): a covariate affected by the confound but not by the policy can identify the effect even with pre-trends (instrument the confound with leads of the policy).

In the Literature: Kleven, Landais, and Søgaard (2019)

The question. How much of the gender gap in earnings is caused by children? Danish registers 1980–2013; parents with a first birth 1985–2003, followed from 5 years before to 10 years after.

The design: an event study around the first birth, separately for mothers and fathers, with age and year dummies:

$$Y_{ist}^g = \sum_{j \neq -1} \alpha_j^g \, \mathbb{1}\{j = t\} + \sum_k \beta_k^g \, \mathbb{1}\{k = \text{age}_{is}\} + \sum_y \gamma_y^g \, \mathbb{1}\{y = s\} + \nu_{ist}^g$$

$$P_t = (\hat{\alpha}_t^m - \hat{\alpha}_t^w) / E[\tilde{Y}_{ist}^w \mid t]$$

with $g \in \{m, w\}$ men or women, t event time, s year, and $\tilde{Y}$ the prediction from age and year alone. P_t is the **child penalty**: the gap that opens, as a share of women's counterfactual earnings.

- No control group of non-parents: identification is smoothness, the same argument as RD in time, with fathers as a check.

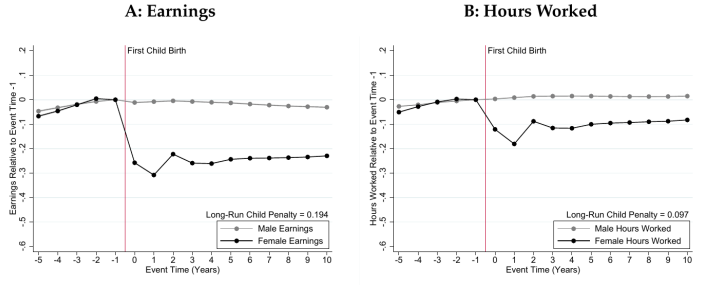
Kleven, Landais, and Søgaard: Reading the Child Penalty

$P_t = (\hat{\alpha}_t^m - \hat{\alpha}_t^w)/E[\tilde{Y}_{ist}^w \mid t]$, step by step, with round numbers (hypothetical, in DKK):

1. Fathers, five years after the first birth: earnings at their $t = -1$ level, so $\hat{\alpha}_5^m = 0$.
2. Mothers: 40,000 below their $t = -1$ level, so $\hat{\alpha}_5^w = -40{,}000$.
3. Gap: $\hat{\alpha}_5^m - \hat{\alpha}_5^w = 0 - (-40{,}000) = 40{,}000$.
4. Scale by what mothers would have earned without children (age and year alone predict 200,000): $P_5 = 40{,}000/200{,}000 = 0.20$.

$P_5 = 0.20$: five years on, children have opened a gap of 20% of women's counterfactual earnings.

Kleven, Landais, and Søgaard: Results



Their Figure 1, panels A–B (accepted manuscript, 2018).

- Mothers' earnings fall almost 30% at birth and settle about 20% below the counterfactual; fathers' do not move. Long-run penalty: earnings 0.194, hours 0.097, participation 0.130, wage rate 0.091.
- The share of gender earnings inequality due to children rose from about 40% (1980) to about 80% (2013).

Staggered

The Setting Everyone Uses ★

Units adopt the policy at *different times*: states pass a law in 1995, 2001, 2008. The standard specification is

$$Y_{it} = \alpha_i + \lambda_t + \beta\, D_{it} + \epsilon_{it}, \qquad D_{it} = \mathbb{1}\{t \geq G_i\}.$$

For thirty years this was assumed to estimate "the" treatment effect. It does when the effect is constant across units and over time since treatment. With heterogeneous or dynamic effects it need not estimate any sensible average.

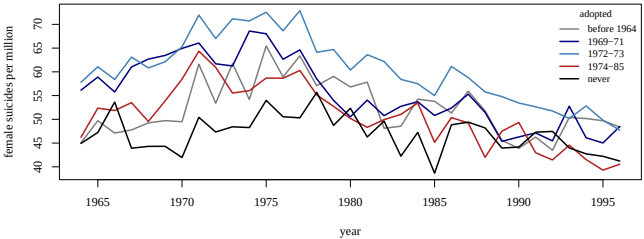
Goodman-Bacon (2021)

The TWFE estimate is a weighted average of *all possible* 2×2 DID comparisons in the data, including comparisons that use already-treated units as controls.

Running Example: Unilateral Divorce and Female Suicide

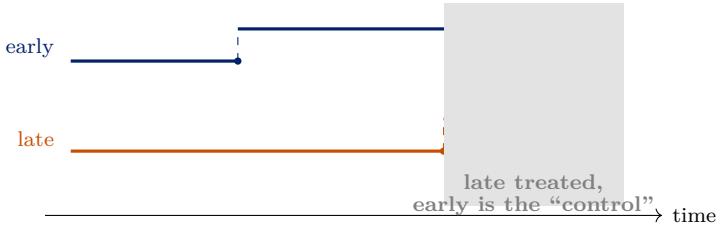
Stevenson and Wolfers (2006): did letting either spouse end a marriage reduce women's suicide? States adopted unilateral divorce in different years.

- 51 states, 1964–96; outcome: female suicides per million women (mean 51 in 1964).
- 9 states adopted before 1964 (always treated), 37 adopted 1969–85, 5 never.



Mean rate by adoption cohort. Their event study (percent change): −9.7% (2.3) averaged over 20 years, −18.7% (3.2) after 19+ years; nothing for men. Data: `bacondecomp::divorce`, crude rates (figs/w10_divorce.R).

The Forbidden Comparison ★



- In the shaded window, TWFE uses the *early-treated* unit as a control for the late-treated one.
- If the early unit's effect is still evolving, that evolution is subtracted from the late unit's effect: it enters with a minus sign.
- With dynamic effects, TWFE can be the wrong *sign* even when every unit-level effect is positive.

The Forbidden Comparison, in Algebra ★

Take the late-versus-early 2×2 from the shaded window: late switches from untreated to treated, early is *treated throughout*. Suppose parallel trends holds exactly and untreated outcomes are flat, so every change in Y is a treatment effect. Then

$$\hat{\beta}^{2\times2}_{\text{late vs. early}} = \underbrace{\text{ATT}_{\text{late}}(\text{post})}_{\text{what we want}} - \underbrace{\Big[\text{ATT}_{\text{early}}(\text{post}) - \text{ATT}_{\text{early}}(\text{pre})\Big]}_{\text{the } \textit{growth} \text{ in early's effect over the window}}.$$

- The first term is what we want: late's effect after it switches.
- The second term is *early's* effect growing over the same window. The comparison treats that growth as if it were a trend, and subtracts it.

The Forbidden Comparison, in Numbers ★

No noise, no trends, every true effect positive:

	$t = 3$	$t = 4$	Change
Late (treated at t=4; effect 1)	0	1	+1
Early (treated at t=2; effects $1, 2, 3$)	2	3	+1
			$\hat{\beta}^{2\times2} = 0$

- Late's true effect is +1; this comparison returns 0. If early's effect grew by 2 per period instead, it would return -1.
- If effects were constant $(1, 1, 1)$, the bracketed term would vanish and the comparison would be harmless, which is exactly why the problem stayed invisible for thirty years of constant-effect thinking.

The Goodman-Bacon Decomposition ★

With staggered timing, the TWFE estimate is a weighted average of every possible 2×2 comparison in the data:

$$\hat{\beta}^{\text{TWFE}} = \sum_{\text{pairs}} s_{ab}\, \hat{\beta}^{2\times 2}_{ab}, \qquad \sum s_{ab} = 1,$$

where the weights s_{ab} depend on group sizes and on the *variance of treatment* within each pair, largest for pairs whose treatment switches mid-window.

Three kinds of comparison arise:

1. Treated versus **never-treated**, clean.
2. **Earlier** treated versus **later** treated, before the later group switches, clean.
3. **Later** treated versus **earlier** treated, after the earlier group switches, the forbidden comparison.

Type 3 uses already-treated units as controls. If their effects are still evolving, that evolution is subtracted from the late group's effect.

The Decomposition on the Divorce Data

TWFE: $\hat{\beta} = -3.05$ (2.32). All 156 2×2 comparisons, grouped (figs/w10_divorce.R; Goodman-Bacon's own numbers, age-adjusted rates, 49 states, in grey):

comparison	weight	average 2×2	weight	average
timing states vs. always treated (pre-1964)	0.41	−6.92	0.38	−7.04
treated vs. never treated	0.23	−5.21	0.24	−5.33
earlier treated vs. later (not yet) treated	0.11	1.25	0.11	−0.19
later treated vs. earlier (already) treated	0.25	3.36	0.26	3.51
TWFE = weighted sum		−3.05		−3.08

- A quarter of the weight sits on comparisons that use already-treated states as controls; their average has the *opposite sign*.
- 41% comes from always-treated states as controls: their whole period is post-treatment, so they are clean only if their effects are constant.

The Goodman-Bacon Weights, Explicitly ★

Let n_k, n_l, n_U be the groups' sample shares and $\bar{D}_j$ the fraction of the panel that group j spends treated (k treated earlier than l). Up to a common normalization, the weight on each 2×2 is the product of the pair's group sizes and the *variance of the treatment dummy within that pair's window*:

$$s_{kU} \propto n_k\, n_U\, \bar{D}_k(1-\bar{D}_k) \qquad \text{treated vs. never-treated}$$

$$s^k_{kl} \propto n_k\, n_l\, (\bar{D}_k - \bar{D}_l)(1-\bar{D}_k) \qquad \text{early vs. late, before late switches}$$

$$s^l_{kl} \propto n_k\, n_l\, \bar{D}_l(\bar{D}_k - \bar{D}_l) \qquad \text{late vs. early, after early switches}$$

- Every weight has the $\bar{D}(1-\bar{D})$ shape of a Bernoulli variance, so groups treated near the middle of the panel dominate: the weights come from OLS mechanics, not from group sizes alone and not from any estimand you chose (Week 1's variance weighting).

The Goodman-Bacon Weights, Explicitly ★ (cont.)

- The forbidden weight s_{kl}^{l} is not negligible in general: it grows with $\bar{D}_k$, i.e. with how long the early group has already been treated when the late group switches.
- All of this is computable from the data alone: `bacondecomp::bacon()` lists every 2×2, its estimate, and its weight. We decompose a real TWFE estimate in lab.

Where the Negative Weights Come From ★

FWL (Week 1) reduces TWFE to a bivariate regression on the *double-demeaned* treatment. With $\bar{D}_i$, $\bar{D}_t$, $\bar{D}$ the unit, period, and grand means of D_{it}:

$$\hat{\beta}^{\text{TWFE}} = \frac{\sum_{it} \tilde{D}_{it}\, Y_{it}}{\sum_{it} \tilde{D}_{it}^2}, \qquad \tilde{D}_{it} = D_{it} - \bar{D}_i - \bar{D}_t + \bar{D}.$$

Substitute $Y_{it} = \alpha_i + \lambda_t + \tau_{it} D_{it} + \epsilon_{it}$: the fixed effects drop out because $\tilde{D}$ has zero unit and period sums, $\sum_{it} \tilde{D}_{it}\epsilon_{it}/NT \to 0$ under parallel trends, and $\sum_{it} \tilde{D}_{it}^2 = \sum_{(i,t):D_{it}=1} \tilde{D}_{it}$, so

$$\text{plim}\ \hat{\beta}^{\text{TWFE}} = \sum_{(i,t):\, D_{it}=1} w_{it}\, \tau_{it}, \qquad w_{it} = \frac{\tilde{D}_{it}}{\sum_{(i,t):\, D_{it}=1} \tilde{D}_{it}}, \qquad \sum w_{it} = 1.$$

Where the Negative Weights Come From ★ (cont.)

- For a *treated* cell, $\tilde{D}_{it} = 1 - \bar{D}_i - \bar{D}_t + \bar{D}$, which is negative exactly when $\bar{D}_i + \bar{D}_t > 1 + \bar{D}$: a unit treated for most of the panel, observed in a period when most of the sample is treated. That is the early adopter, late in the panel.
- Intuition: OLS rewards cells whose treatment status is *surprising* given their unit and period averages. An early adopter's late-period treatment is completely unsurprising, so its effect is priced at zero, or below zero.
- The weights are a mechanical fact about OLS and are there even under constant effects, harmlessly. Heterogeneous τ_{it} is what turns strange weights into a wrong answer.

Negative Weights: Consequences and Diagnostics ★

de Chaisemartin and D'Haultfœuille (2020) generalize the previous frame's calculation, allowing covariates and non-binary treatments, and show that negative weights are the rule in staggered designs, not a corner case.

- Consequence: TWFE can have the wrong sign even when every $\tau_{it} > 0$. This is not a small-sample problem; it survives $n \to \infty$, because the weights, not the noise, are at fault.
- By the formula, the negative weights fall on units treated *early* and observed over a *long* post-period, exactly the units carrying the most information about long-run effects.

Negative Weights: Consequences and Diagnostics ★ (cont.)

- **Diagnostic**: compute the weights (`TwoWayFEWeights::twowayfeweights`) and report how many are negative and what share of the total mass they carry. Small negative mass is reassuring about *this particular* problem; it says nothing about parallel trends or about dynamic-effect contamination, so it is one diagnostic, not a validity test.
- Sun and Abraham (2021) show the same contamination in event-study coefficients: with staggered timing, the TWFE $\hat{\tau}_k$ at lag k is the cohorts' effects at lag k (weights sum to one) *plus* their effects at other lags (weights sum to zero, some negative). So apparent pre-trends can be artefacts of post-treatment dynamics, and a clean pre-trend plot from raw TWFE proves nothing.

TWFE is fine when treatment timing is common, or when effects are genuinely constant. It is unreliable exactly where sociologists use it most.

Negative Weights on the Divorce Data

The formula two frames back, computed cell by cell: $w_{it} = \tilde{D}_{it} / \sum_{D_{it}=1} \tilde{D}_{it}$ with $\tilde{D}$ the double-demeaned treatment (figs/w10_divorce.R).

treated state-years	1,164
with a negative weight	267
sum of the negative weights	−0.38
states carrying them	adopted before 1964, 1969, 1970
years	1974–1996

- Exactly where the formula said: early adopters, late in the panel. Their long-run effects, which Stevenson and Wolfers find are the largest, enter TWFE with the wrong sign.
- Positive weights sum to 1.38, so TWFE is not a convex average of any group's effect.

Callaway–Sant'Anna: Build Clean Blocks

Define the group-time average treatment effect for cohort g at time t:

$$ATT(g,t) = E\big[Y_t(g) - Y_t(\infty) \mid G_i = g\big],$$

identified by comparing cohort g to units *not yet treated* at t (or to never-treated units), using only periods before g as the baseline. Under cohort-level parallel trends and no anticipation,

$$ATT(g,t) = E\big[Y_t - Y_{g-1} \mid G_i = g\big] - E\big[Y_t - Y_{g-1} \mid \text{not yet treated at } t\big] :$$

a clean 2×2. The baseline $g-1$ predates cohort g's treatment, the comparison group has not started treatment by t, and no already-treated unit is ever used as a control.

Toy example (the forbidden-comparison numbers, plus a never-treated unit with $Y = 0$): late's block at $t = 4$ is $ATT(4,4) = (1-0) - (0-0) = 1$, its true effect. The early unit is never used as a control (figs/w10_toy_staggered.R).

Callaway–Sant'Anna: Then Aggregate

- Under *conditional* parallel trends, each block can be estimated by IPW, outcome regression, or a doubly robust estimator, Week 6, reappearing inside a DID.
- **Then choose an aggregation**, and the choice is an estimand choice:
  - *simple*: average over all treated (g, t);
  - *dynamic*: average by $t - g$, giving an event-study path;
  - *group*: average within cohort, showing who benefited;
  - *calendar*: average within period, showing when.
- Sun–Abraham reaches the same place through an interaction-weighted regression; Borusyak–Jaravel–Spiess by imputing $Y(0)$ from untreated cells.

Imputation, Step by Step (Borusyak, Jaravel, and Spiess 2024)

Use only untreated cells to learn the untreated model, then fill in:

1. $Y_{it} = \alpha_i + \lambda_t + \epsilon_{it}$ on cells with $D_{it} = 0$ — never-treated, and pre-periods of the treated
2. $\hat{Y}_{it}(0) = \hat{\alpha}_i + \hat{\lambda}_t$ for cells with $D_{it} = 1$ — impute the missing potential outcome
3. $\hat{\tau}_{it} = Y_{it} - \hat{Y}_{it}(0), \quad \hat{\tau} = \sum_{D_{it}=1} v_{it}\,\hat{\tau}_{it}$ — average with weights you choose

- No treated cell is ever a control; any aggregation (v_{it}) is allowed, so the estimand is chosen, not inherited from OLS.
- **Efficient** when errors are homoskedastic and serially uncorrelated: it uses all pre-periods, not just $g - 1$.
- **Pre-trend tests are separate**: run on untreated cells only, so testing does not change the estimate.
- Gardner (2022)'s two-stage DiD is the same idea as two regressions (`did2s`).

Imputation on the Toy Example

Untreated outcome 0 everywhere; bold cells are treated:

Y_{it}	$t=1$	$t=2$	$t=3$	$t=4$
never treated	0	0	0	0
early (from $t=2$; effects 1, 2, 3)	0	**1**	**2**	**3**
late (from $t=4$; effect 1)	0	0	0	**1**

1. Fit $\alpha_i + \lambda_t$ on the 8 untreated cells. All are 0, so every $\hat{\alpha}_i = \hat{\lambda}_t = 0$.
2. Impute $\hat{Y}_{it}(0) = 0$ in the 4 treated cells.
3. $\hat{\tau}_{it} = Y_{it} - 0$: 1, 2, 3 and 1. Average: $7/4 = 1.75$, the true average effect on the treated cells.

TWFE on the same 12 cells: 1.10; without the never-treated unit: 0.50 (figs/w10_toy_staggered.R).

Extended TWFE: the Regression Was Not the Problem

(Wooldridge 2025)

TWFE failed because it forced one β on all treated cells. Give every cohort–period its own coefficient:

$$Y_{it} = \alpha_i + \lambda_t + \sum_g \sum_{t \geq g} \tau_{gt} \, \mathbb{1}\{G_i = g\} \, \mathbb{1}\{\text{period} = t\} + \epsilon_{it}.$$

- The unit and period effects are then fitted on untreated cells only (every treated cell has its own dummy), so $\hat{\tau}_{gt}$ equals the imputation estimate. Divorce data: both give -4.85.
- Adding the pre-period interactions too, against never-treated states, reproduces Sun–Abraham and Callaway–Sant'Anna with a universal base: all three give -10.56.
- Covariates enter as cohort-by-period interactions (the "two-way Mundlak" regression). Nonlinear versions (logit, Poisson) follow (`etwfe`).

Stacked DiD and Its Weights

Stacking (Cengiz, Dube, Lindner, and Zipperer 2019): for each adoption cohort, build a sub-experiment (the cohort plus clean controls, in a fixed window around adoption), stack them, and run one regression with sub-experiment-specific unit and time effects.

- Never uses already-treated controls, and is easy to explain.
- Unweighted stacking does not estimate a clean average: each sub-experiment's treated and control shares differ, so OLS weights their trends unequally (Wing, Freedman, and Hollingsworth 2024). Corrective sample weights recover a "trimmed aggregate ATT".
- Used in sociology: Walmart Supercenters and unions (Choper, Lehner, and Parolin 2026); welfare reform and children's home environments (Kalil et al. 2023), both later.

The Modern Estimators ★

All of them share one principle: never use an already-treated unit as a control. Build clean 2×2 comparisons, then aggregate with weights you choose.

Method	Idea	R
Callaway–Sant'Anna (2021)	ATT(g, t); never- or not-yet-treated controls	`did`
Sun–Abraham (2021)	interaction-weighted event study	`fixest::sunab`
de Chaisemartin–D'Haultfœuille	clean comparisons at switch dates	`DIDmultiplegt`
Borusyak–Jaravel–Spiess	impute $Y(0)$ from untreated cells	`didimputation`
Stacked / cohort DID	one clean event window per cohort	manual

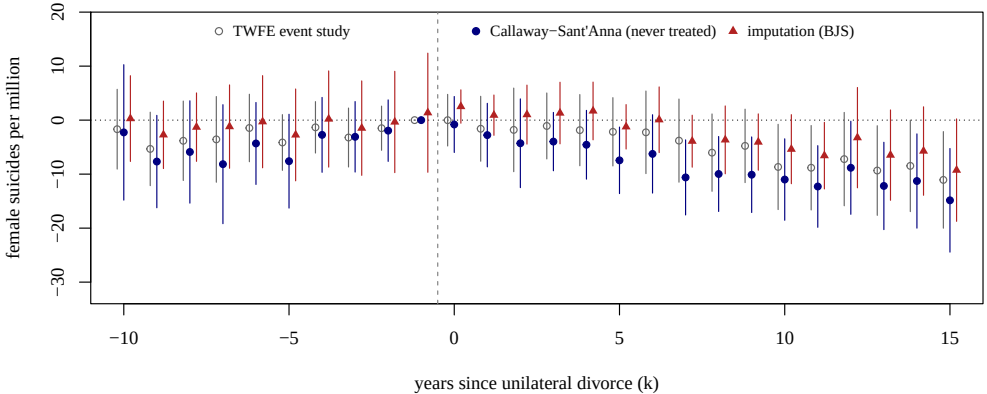
Six Estimators, One Dataset

Divorce data without the 9 always-treated states (they have no pre-period); standard errors clustered by state (figs/w10_divorce.R):

estimator	controls, baseline	ATT	(se)
TWFE, all 51 states	everything	-3.05	(2.32)
TWFE, 42 states	everything	-0.34	(2.76)
Callaway–Sant'Anna	never treated, year $g-1$	-10.56	(3.12)
Callaway–Sant'Anna	not yet treated, year $g-1$	-8.69	(3.40)
Sun–Abraham	never treated, year $g-1$	-10.56	(3.52)
imputation = extended TWFE	not yet treated, all pre-years	-4.85	(2.97)

- The robust estimators agree on the sign and say TWFE understates the effect.
- They differ in size because they assume parallel trends over different windows ($g-1$ only, or every pre-period) and weight the (g,t) cells differently.

The Event Studies, Side by Side



- Leads are not flat (Callaway–Sant'Anna at $k = -5$: -7.6): untreated trends were not parallel in every pre-year, so "which pre-years define the baseline" changes the answer.
- Effects grow with time since adoption (-14.9 at $k = 15$), the dynamic heterogeneity that breaks TWFE.

Choices Every Staggered DiD Makes

choice	options	consequence
comparison group	never treated; not yet treated	not-yet uses more data but needs no anticipation and parallel trends with future adopters
baseline	$g-1$ only; all pre-periods	all pre-periods is efficient if trends are parallel throughout, biased if not
aggregation	simple; by event time; by cohort; by calendar time	each is a different estimand; say which
anticipation	shift g back by δ periods	otherwise effects leak into the baseline
covariates	none; DR within each $ATT(g,t)$	use pre-treatment values only
inference	cluster; bootstrap; randomization	few treated cohorts need care

Efficiency when adoption timing is as good as random: Roth and Sant'Anna (2023b) (`staggered`). A design-based reading of TWFE under random timing: Athey and Imbens (2022).

Honest Inference About Pre-Trends

Rather than testing pre-trends and proceeding as if they had passed, Rambachan and Roth (2023) propose bounding how badly parallel trends could fail.

- Their "relative magnitudes" restriction: after treatment, the violation of parallel trends may change from one period to the next by at most $\bar{M}$ times the largest such change before treatment. The confidence sets account for sampling error in the pre-period estimates.
- Report the *breakdown value*: how large $\bar{M}$ would have to be before your conclusion reverses.
- This turns an untestable assumption into a quantitative claim, exactly the move sensitivity analysis made in Week 6.
- `HonestDiD`.

Honest Inference About Pre-Trends (cont.)

Inference reminder

Bertrand, Duflo, and Mullainathan (2004): cluster on the unit at which treatment varies. With few clusters, use CR2 or a wild cluster bootstrap (Week 2).

Sensitivity on the Divorce Data

Callaway–Sant'Anna event study, universal base; target: the average effect over $k = 0, \ldots, 10$, estimated at -6.53 (figs/w10_divorce.R, `HonestDiD`).

allowed post-period violation, relative to the largest pre-period one ($\bar{M}$)	95% interval
none (parallel trends exact)	$[-11.9, -1.1]$
$\bar{M} = 0.05$	$[-13.4, 0.8]$
$\bar{M} = 0.25$	$[-28.4, 16.4]$
$\bar{M} = 1$	$[-55.1, 55.1]$

- Breakdown below $\bar{M} = 0.05$: allowing post-period violations one-twentieth the size of the largest pre-period swing erases the sign.
- The pre-period swings are large (leads near -7.6), so relative magnitudes give little room.

Beyond

Doses: a Continuous Treatment

Callaway, Goodman-Bacon, and Sant'Anna (2025). Units receive dose d (minimum-wage increase in dollars, hours of a program); some get $d = 0$. Two different questions:

$$\mathrm{ATT}(a \mid d) = E[Y_t(a) - Y_t(0) \mid D = d] \qquad \text{effect of dose } a \text{ on those who got } d$$

$$\mathrm{ATT}(d \mid d) - \mathrm{ATT}(d' \mid d') = \underbrace{\mathrm{ATT}(d \mid d) - \mathrm{ATT}(d' \mid d)}_{\text{causal: more dose, same units}} + \underbrace{\mathrm{ATT}(d' \mid d) - \mathrm{ATT}(d' \mid d')}_{\text{selection: different units}}$$

- Ordinary parallel trends (each dose group vs. $d = 0$) identifies each $\mathrm{ATT}(d \mid d)$, but not how effects change with the dose.
- Comparing doses needs **strong parallel trends**: every dose group would have trended like any other under every dose. It rules out selection on gains.
- TWFE on a continuous D mixes both terms with weights that do not follow the dose distribution; its weights on level effects sum to zero (`contdid`).

Treatments That Switch On and Off

Union membership, unemployment, living with a partner: units enter and leave treatment, and past treatment may still matter.

- TWFE with D_{it} switching assumes no carryover (Imai–Kim) and constant effects, and uses switchers as each other's controls.
- de Chaisemartin and D'Haultfœuille (2026): compare units whose treatment first changes at t with units that had the *same* treatment history up to t and have not yet changed. The effect of having been exposed for ℓ periods, DID_ℓ, is an event study for non-absorbing, non-binary treatments (`DIDmultiplegtDYN`).
- Matching on treatment histories (Imai, Kim, and Wang 2023) does the same with explicit lag length L.

Cost: only units with a matching history count, so fewer comparisons and wider intervals. That is the price of not assuming carryover away.

Triple Differences

A policy hits one group (say, mothers) in treated states. Difference the state DiD for mothers against the state DiD for a group the policy should not touch (women without children):

$$\text{DDD} = \text{DiD}_{\text{mothers}} - \text{DiD}_{\text{non-mothers}}.$$

- Needs only **one** parallel-trends assumption: the bias in the two DiDs is the same. Neither DiD needs to be unbiased on its own (Olden and Møen 2022).
- With covariates or staggered timing, "the difference of two DiDs" and three-way fixed effects regressions are generally not valid; pooling not-yet-treated units can add bias even without covariates. Use estimators built for DDD (Ortiz-Villavicencio and Sant'Anna 2025; `tripledifff`).

Distributions, and a Design-Based View

Effects on the distribution (Callaway and Li 2019): with three or more panel periods, distributional parallel trends plus copula stability (the link between a unit's change in $Y(0)$ and its level stays the same over time) identify quantile effects on the treated.

Where does uncertainty come from? Standard errors here usually imagine resampling units. In policy panels we observe every state.

- Athey and Imbens (2022): treat the *adoption date* as the random element; TWFE then has a design-based reading, and the usual variance is conservative.
- Rambachan and Roth (2026): design-based uncertainty for quasi-experiments; usual standard errors can remain valid, if conservative, for a well-defined estimand.

Factor models (unit-specific responses to common shocks) relax parallel trends further: synthetic DiD, generalized synthetic control, counterfactual estimators (`fect`). Next week.

Sociology

In the Literature: Harknett, Schneider, and Irwin (2021)

The question. Does making work schedules predictable improve workers' lives? Seattle's Secure Scheduling Ordinance (2017): two weeks' notice, pay for last-minute changes, extra pay for "clopening" shifts; large retail and food-service firms.

The data. The Shift Project: surveys of hourly workers recruited by Facebook and Instagram ads. Seattle: 754 workers at baseline (spring 2017), 747 after one year, 441 after two. Comparison: workers at the *same firms* in 24 large metros with above-federal minimum wages. 17,689 responses in all.

The design. DiD with two post periods,

$$Y_{ict} = \beta_1 \,\text{Seattle}_c \times \text{Year1}_t + \beta_2 \,\text{Seattle}_c \times \text{Year2}_t + \text{city and wave effects} + X'_{ict}\delta + \epsilon_{ict},$$

linear probability models, weighted to the American Community Survey. Same firms on both sides narrows differences in business model and jobs, as Card and Krueger held the industry fixed.

Harknett, Schneider, and Irwin: Checking the Design

Baseline levels	Similar: unpredictability scale 2.79 vs. 2.85; happy 76% vs. 75%; any material hardship 60% vs. 58%. Clopening differs (37% vs. 44%).
Pre-trends	No pre-2017 data, so no event study. Instead: in places without scheduling laws, the national scale stays near 2.8 in 2017, 2018, 2019.
Comparison group	Results hold with cities that considered but did not pass a law, with a propensity-matched sample, and (smaller) with workers just outside Seattle, where spillover is plausible.
Composition	The Seattle sample becomes slightly younger and shorter-tenured.
Sampling	Not a probability sample; more women and White workers than the workforce.

A one-treated-city DiD: inference rests on the comparison cities behaving like Seattle would have. Next week's synthetic control is built for exactly this case.

Harknett, Schneider, and Irwin: Main Results

Change in Seattle minus change in comparison cities after the law (their Figures 1–2):

outcome	DiD estimate
schedule unpredictability scale (baseline 2.8)	-0.50 (-18%)
at least two weeks' notice	$+11$ pts
last-minute changes without pay	-13 pts
on-call shifts	-7 pts $(p \leq 0.07)$
happy	$+7$ pts
good sleep quality	$+11$ pts
any material hardship	-10 pts
psychological distress	n.s.

The law moved schedules and well-being. An IV extension uses the law as an instrument for predictability (first-stage $F = 17.1$).

Staggered DiD in Sociology: Two Recent Studies

Choper, Lehner, and Parolin (2026, ASR): Walmart Supercenters and unions. Restricted PSID 1983–2019; 966 first Supercenter openings across counties; stacked DiD with never-treated controls, matched workers, ±10 years.

- Union membership falls 1.4 points (se 0.8); 3.5 points (1.3) when differential linear pre-trends are removed: about 21% of the 16.1% baseline.
- The headline depends on a trend adjustment: report both, as they do.

Kalil, Corman, Dave, Schwartz-Soicher, and Reichman (2023, *Demography*): staggered state welfare reform 1992–98; NLSY79 unmarried, low-educated mothers of young children.

home environment score	TWFE	stacked (not-yet-treated controls)
emotional support	−4.95 (1.44)	−6.56 (1.77)
overall	−2.89	−5.40

Already-treated controls attenuate TWFE when effects grow, as in the divorce data.

Practice

What a Reanalysis of 49 Studies Found (Chiu, Lan, Liu, and Xu 2026)

Political scientists reran 49 published panel studies with the heterogeneity-robust estimators of this lecture.

- The new estimators usually agree with TWFE *qualitatively*, but the estimates vary a lot across estimators.
- Many studies lack evidence for parallel trends once pre-periods are examined properly.
- Many are underpowered: the robust estimators' intervals are wide.

Read with the divorce data in mind: the estimator mattered (TWFE vs. robust), and so did the window over which trends are assumed parallel. The practical guide that collects these steps: Baker, Callaway, Cunningham, Goodman-Bacon, and Sant'Anna (2026).

A DID Checklist ★

1. State the estimand: ATT for whom, over which post-treatment window?
2. Plot the raw group means over time. Before any regression.
3. Date treatment and defend no anticipation; if in doubt, shift G_i earlier.
4. Is timing staggered? If so, do not use plain TWFE. Use a modern estimator, and report the TWFE estimate only for comparison.
5. Report an event-study plot with pre-period coefficients, using Sun–Abraham or Callaway–Sant'Anna rather than raw TWFE leads.
6. Justify the scale (levels or logs) on substantive grounds.
7. Cluster on the treatment-assignment unit; report the number of clusters.
8. Report a sensitivity analysis for parallel trends.
9. Placebo: a fake treatment date, or an outcome the policy should not affect.

Looking Ahead

- Fixed effects difference away time-invariant confounders, bought with precision and with strict exogeneity; DiD adds parallel trends.
- The staggered-adoption literature is a case study in how a method can be standard for decades and still be wrong. Read papers published before 2021 with this in mind, including your own field's classics.
- **Next week**: what to do when there is only *one* treated unit, and "the control group" has to be constructed.
- Notice that conditional parallel trends brought Week 6 back inside a DID: once trends are only parallel given covariates, you need a weighting or outcome model again, and all of Week 6's cautions return with it.

Problem Set 4

Assigned today, due Monday November 9. Covers Weeks 7 and 9.

References

- Abadie, Alberto. 2005. "Semiparametric Difference-in-Differences Estimators." *Review of Economic Studies* 72(1): 1–19.
- Anderson, T. W. and Cheng Hsiao. 1981. "Estimation of Dynamic Models with Error Components." *Journal of the American Statistical Association* 76(375): 598–606.
- Angrist, Joshua D. and Jörn-Steffen Pischke. 2009. *Mostly Harmless Econometrics*. Princeton University Press. [Ch. 5]
- Arellano, Manuel and Stephen Bond. 1991. "Some Tests of Specification for Panel Data: Monte Carlo Evidence and an Application to Employment Equations." *Review of Economic Studies* 58(2): 277–297.
- Athey, Susan and Guido W. Imbens. 2022. "Design-Based Analysis in Difference-in-Differences Settings with Staggered Adoption." *Journal of Econometrics* 226(1): 62–79.
- Baker, Andrew, Brantly Callaway, Scott Cunningham, Andrew Goodman-Bacon, and Pedro H.C. Sant'Anna. 2026. "Difference-in-Differences Designs: A Practitioner's Guide." *Journal of Economic Literature* 64(2): 498–557.

References (cont.)

- Bertrand, Marianne, Esther Duflo, and Sendhil Mullainathan. 2004. "How Much Should We Trust Differences-in-Differences Estimates?" *Quarterly Journal of Economics* 119(1): 249–275.
- Borusyak, Kirill, Xavier Jaravel, and Jann Spiess. 2024. "Revisiting Event-Study Designs: Robust and Efficient Estimation." *Review of Economic Studies* 91(6): 3253–3285.
- Brüderl, Josef and Volker Ludwig. 2015. "Fixed-Effects Panel Regression." Pp. 327–357 in *The SAGE Handbook of Regression Analysis and Causal Inference*, edited by H. Best and C. Wolf. Sage.
- Budig, Michelle J. and Paula England. 2001. "The Wage Penalty for Motherhood." *American Sociological Review* 66(2): 204–225.
- Caetano, Carolina and Brantly Callaway. 2024. "Difference-in-Differences when Parallel Trends Holds Conditional on Covariates." arXiv:2406.15288.
- Callaway, Brantly, Andrew Goodman-Bacon, and Pedro H.C. Sant'Anna. 2025. "Difference-in-Differences with a Continuous Treatment." arXiv:2107.02637 (NBER WP 32117).

References (cont.)

- Callaway, Brantly and Tong Li. 2019. "Quantile Treatment Effects in Difference in Differences Models with Panel Data." *Quantitative Economics* 10(4): 1579–1618.
- Callaway, Brantly and Pedro H.C. Sant'Anna. 2021. "Difference-in-Differences with Multiple Time Periods." *Journal of Econometrics* 225(2): 200–230.
- Card, David and Alan B. Krueger. 1994. "Minimum Wages and Employment: A Case Study of the Fast-Food Industry in New Jersey and Pennsylvania." *American Economic Review* 84(4): 772–793.
- Cengiz, Doruk, Arindrajit Dube, Attila Lindner, and Ben Zipperer. 2019. "The Effect of Minimum Wages on Low-Wage Jobs." *Quarterly Journal of Economics* 134(3): 1405–1454.
- Chiu, Albert, Xingchen Lan, Ziyi Liu, and Yiqing Xu. 2026. "Causal Panel Analysis under Parallel Trends: Lessons from a Large Reanalysis Study." *American Political Science Review* 120(1): 245–266.
- Choper, Joshua, Clemens Lehner, and Zachary Parolin. 2026. "Countervailing Powers: Labor Unions Against the Buyer Power of Walmart Supercenters." *American Sociological Review* 91(3): 435–463.

References (cont.)

- de Chaisemartin, Clément and Xavier D'Haultfœuille. 2020. "Two-Way Fixed Effects Estimators with Heterogeneous Treatment Effects." *American Economic Review* 110(9): 2964–2996.
- de Chaisemartin, Clément and Xavier D'Haultfœuille. 2026. "Difference-in-Differences Estimators of Intertemporal Treatment Effects." *Review of Economics and Statistics* 108(4): 863–880.
- Freyaldenhoven, Simon, Christian Hansen, Jorge Pérez Pérez, and Jesse M. Shapiro. 2025. "Visualization, Identification, and Estimation in the Linear Panel Event-Study Design." Pp. 225–268 in *Advances in Economics and Econometrics: Twelfth World Congress*, Vol. 2. Cambridge University Press.
- Freyaldenhoven, Simon, Christian Hansen, and Jesse M. Shapiro. 2019. "Pre-Event Trends in the Panel Event-Study Design." *American Economic Review* 109(9): 3307–3338.
- Gardner, John. 2022. "Two-Stage Differences in Differences." arXiv:2207.05943.
- Goodman-Bacon, Andrew. 2021. "Difference-in-Differences with Variation in Treatment Timing." *Journal of Econometrics* 225(2): 254–277.

References (cont.)

- Harknett, Kristen, Daniel Schneider, and Véronique Irwin. 2021. "Improving Health and Economic Security by Reducing Work Schedule Uncertainty." *Proceedings of the National Academy of Sciences* 118(42): e2107828118.
- Heckman, James J., Hidehiko Ichimura, and Petra E. Todd. 1997. "Matching as an Econometric Evaluation Estimator: Evidence from Evaluating a Job Training Programme." *Review of Economic Studies* 64(4): 605–654.
- Hausman, Jerry A. 1978. "Specification Tests in Econometrics." *Econometrica* 46(6): 1251–1271.
- Imai, Kosuke and In Song Kim. 2019. "When Should We Use Unit Fixed Effects Regression Models for Causal Inference with Longitudinal Data?" *American Journal of Political Science* 63(2): 467–490.
- Imai, Kosuke, In Song Kim, and Erik H. Wang. 2023. "Matching Methods for Causal Inference with Time-Series Cross-Sectional Data." *American Journal of Political Science* 67(3): 587–605.
- Kalil, Ariel, Hope Corman, Dhaval M. Dave, Ofira Schwartz-Soicher, and Nancy E. Reichman. 2023. "Welfare Reform and the Quality of Young Children's Home Environments." *Demography* 60(6): 1791–1813.

References (cont.)

- Kleven, Henrik, Camille Landais, and Jakob Egholt Søgaard. 2019. "Children and Gender Inequality: Evidence from Denmark." *American Economic Journal: Applied Economics* 11(4): 181–209.
- Ludwig, Volker and Josef Brüderl. 2018. "Is There a Male Marital Wage Premium? New Evidence from the United States." *American Sociological Review* 83(4): 744–770.
- Nickell, Stephen. 1981. "Biases in Dynamic Models with Fixed Effects." *Econometrica* 49(6): 1417–1426.
- Olden, Andreas and Jarle Møen. 2022. "The Triple Difference Estimator." *Econometrics Journal* 25(3): 531–553.
- Ortiz-Villavicencio, Marcelo and Pedro H.C. Sant'Anna. 2025. "Better Understanding Triple Differences Estimators." arXiv:2505.09942.
- Rambachan, Ashesh and Jonathan Roth. 2023. "A More Credible Approach to Parallel Trends." *Review of Economic Studies* 90(5): 2555–2591.

References (cont.)

- Rambachan, Ashesh and Jonathan Roth. 2026. "Design-Based Uncertainty for Quasi-Experiments." *Journal of the American Statistical Association* 121(553): 477–491.
- Roodman, David. 2009. "A Note on the Theme of Too Many Instruments." *Oxford Bulletin of Economics and Statistics* 71(1): 135–158.
- Roth, Jonathan. 2022. "Pretest with Caution: Event-Study Estimates after Testing for Parallel Trends." *American Economic Review: Insights* 4(3): 305–322.
- Roth, Jonathan. 2026. "Interpreting Event-Studies from Recent Difference-in-Differences Methods." *Japanese Economic Review* 77(2): 275–288.
- Roth, Jonathan and Pedro H.C. Sant'Anna. 2023a. "When Is Parallel Trends Sensitive to Functional Form?" *Econometrica* 91(2): 737–747.
- Roth, Jonathan and Pedro H.C. Sant'Anna. 2023b. "Efficient Estimation for Staggered Rollout Designs." *Journal of Political Economy Microeconomics* 1(4): 669–709.
- Roth, Jonathan, Pedro H.C. Sant'Anna, Alyssa Bilinski, and John Poe. 2023. "What's Trending in Difference-in-Differences?" *Journal of Econometrics* 235(2): 2218–2244.

References (cont.)

- Rüttenauer, Tobias and Volker Ludwig. 2023. "Fixed Effects Individual Slopes: Accounting and Testing for Heterogeneous Effects in Panel Data or Other Multilevel Models." *Sociological Methods & Research* 52(1): 43–84.
- Sant'Anna, Pedro H.C. and Jun Zhao. 2020. "Doubly Robust Difference-in-Differences Estimators." *Journal of Econometrics* 219(1): 101–122.
- Stevenson, Betsey and Justin Wolfers. 2006. "Bargaining in the Shadow of the Law: Divorce Laws and Family Distress." *Quarterly Journal of Economics* 121(1): 267–288.
- Sun, Liyang and Sarah Abraham. 2021. "Estimating Dynamic Treatment Effects in Event Studies with Heterogeneous Treatment Effects." *Journal of Econometrics* 225(2): 175–199.
- Vella, Francis and Marno Verbeek. 1998. "Whose Wages Do Unions Raise? A Dynamic Model of Unionism and Wage Rate Determination for Young Men." *Journal of Applied Econometrics* 13(2): 163–183.
- Wing, Coady, Seth M. Freedman, and Alex Hollingsworth. 2024. "Stacked Difference-in-Differences." NBER Working Paper 32054.

References (cont.)

- Wooldridge, Jeffrey M. 2010. *Econometric Analysis of Cross Section and Panel Data*. 2nd ed. MIT Press.
- Wooldridge, Jeffrey M. 2025. "Two-Way Fixed Effects, the Two-Way Mundlak Regression, and Difference-in-Differences Estimators." *Empirical Economics* 69(5): 2545–2587.